

MODULE 2.3

Network Hardening: Services and Protocols

Network Defense | Cisco Networking Academy

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Topics Covered:

- 2.3.1 • Network Vulnerabilities & Hardening Overview
- 2.3.2 • Network and Routing Services — DHCP, DNS, ICMP, RIP, NTP
- 2.3.3 • Telnet, SSH and SCP
- 2.3.4 • Secure Protocols — SNMP, HTTP/S, FTP, POP, IMAP, MIME

2.3.1 Network Vulnerabilities & Hardening

Why hardening networks is essential for protecting @Apollo

What is Network Hardening?

- Network vulnerabilities open to attack and expose the organisation and its customers.
- Network hardening reduces the attack surface by eliminating vulnerabilities in services and protocols.
- Only necessary services and ports should be running and exposed on network devices.
- Cybercriminals exploit vulnerable network services to attack devices or launch larger attacks.

Port Scanning

- A port scanner sends a message to each port and waits for a response to determine how it is used.
- Both security professionals and cybercriminals use port scanners — for defence and attack.
- Securing services ensures that only necessary ports are exposed and available.

2.3.2 Network Services — DHCP

Dynamic Host Configuration Protocol

What is DHCP?

- DHCP uses a server to assign IP addresses and network configuration information to devices.
- Attackers can target DHCP servers to deny devices access to the network.
- DHCP Snooping prevents rogue DHCP servers from providing IP addresses to clients.

DHCP Security Checklist

- Physically secure the DHCP server
- Apply all software patches promptly
- Locate the DHCP server behind a firewall
- Monitor DHCP activity by reviewing logs regularly
- Maintain a strong antivirus solution on the server
- Uninstall any unused services and applications
- Close all unused ports on the server

2.3.2 Network Services — DNS

Domain Name System

What is DNS?

- DNS translates domain names such as `www.cisco.com` into numerical IP addresses.
- Attackers can target DNS servers to redirect traffic to rogue websites or deny access to resources.
- DNSSEC (DNS Security Extensions) uses digital signatures to strengthen authentication.

DNS Security Checklist

- Keep DNS software up to date at all times
- Prevent version strings from revealing server information
- Separate internal and external DNS servers
- Restrict allowed transactions by client IP address
- Use transaction signatures to authenticate DNS transactions
- Disable or restrict zone transfers and dynamic updates
- Enable logging and regularly analyse the logs
- Implement DNSSEC and sign all zones

2.3.2 Network Services — ICMP, RIP & NTP

Routing and messaging protocols that require hardening

ICMP — Internet Control Messaging Protocol

- Used by network devices to send error messages when a service or host is unreachable.
- Cybercriminals misuse ICMP for reconnaissance, Denial of Service and covert channel attacks.
- Countermeasure: Filter ICMP requests on the network to prevent misuse.

RIP — Routing Information Protocol

- RIP limits network paths to a maximum of 15 hops and calculates the best route by hop count.
- Attacks can degrade performance, cause service outages or redirect traffic to rogue paths.
- Countermeasure: Use authentication for routing updates and apply regular patches.

NTP — Network Time Protocol

- NTP synchronises clocks across network devices. Correct timestamps are critical for security logs, syslog data and digital certificate validation.
- Attackers target time servers to disrupt secure communications and hide evidence of attacks.
- Countermeasure: Use NTP Authentication to verify that the server is trusted.

2.3.3 Telnet vs SSH

Comparing insecure and secure remote access protocols

Telnet — TCP Port 23 (Insecure)

- Telnet is an older protocol that uses plaintext when authenticating (username and password).
- All data transmitted over a Telnet session is in plaintext and can be intercepted.
- Cybercriminals use tools like Wireshark to capture credentials from Telnet sessions.

SSH — TCP Port 22 (Secure — Recommended)

- SSH provides a secure, encrypted remote connection to a device.
- All communication including usernames and passwords is encrypted, preventing eavesdropping.
- SSH should always be used instead of Telnet for managing network device connections.
- In a Wireshark capture of an SSH session, the data appears encrypted — only IP addresses are visible.

SCP — Secure Copy Protocol

- SCP securely transfers files between two remote systems using SSH for transfer and authentication.
- SCP ensures the authenticity, confidentiality and integrity of files in transit.

2.3.4 Secure Protocols — SNMP & HTTP/S

Replacing insecure protocols to protect network infrastructure

SNMP — Simple Network Management Protocol

- SNMP collects statistics from TCP/IP devices to monitor network and computer equipment.
- SNMPv3 is the current standard — it uses cryptography to prevent eavesdropping and ensures data integrity during transit.
- Older versions (v1 and v2c) lack strong security and should be avoided or disabled.

HTTP vs HTTPS

- HTTP (port 80) transmits all content in plaintext, leaving users open to monitoring and attack.
- SSL (Secure Sockets Layer) uses a handshake at the start of a session to encrypt communication and prevent eavesdropping and tampering.
- TLS (Transport Layer Security) is the updated, more secure replacement for SSL.
- When SSL/TLS is in use, the browser shows HTTPS instead of HTTP in the address bar.
- Always use HTTPS — never use plain HTTP for sensitive communication.

2.3.4 Secure Protocols — FTP & Email

Securing file transfers and email communications

FTP vs FTPS

- FTP (File Transfer Protocol) transfers files using a plaintext username and password that can easily be intercepted by an attacker.
- FTPS (FTP Secure) adds TLS and SSL support to FTP, preventing eavesdropping, tampering and forgery on all exchanged messages.
- Always use FTPS or SFTP (FTP over SSH) — never use plain FTP.

Email — POP, IMAP, MIME & S/MIME

- Email uses POP (port 110) and IMAP (port 143) to retrieve messages, and MIME to attach non-text content such as images and documents.
- To secure POP and IMAP, use SSL/TLS to encrypt email during transmission.
- S/MIME (Secure/Multipurpose Internet Mail Extensions) sends digitally signed and encrypted messages.
- S/MIME provides authentication, message integrity and non-repudiation.

2.3 Summary — Network Hardening: Services and Protocols

Key concepts from this module

Services & Remote Access

- DHCP — Use DHCP Snooping to block rogue servers
- DNS — Implement DNSSEC for authentication
- ICMP — Filter requests to prevent DoS attacks
- RIP — Use authentication and apply patches
- NTP — Use NTP Authentication to verify servers

- SSH (Port 22) — Always use instead of Telnet
- SCP — Use for secure file transfers over SSH

Secure Protocol Standards

- SNMPv3 — Encrypted network monitoring
- TLS / SSL — Encrypts HTTP, FTP and email
- HTTPS — Always use over plain HTTP
- FTPS — Always use over plain FTP
- DNSSEC — Protects DNS integrity
- S/MIME — Signed and encrypted email
- NTP Auth — Verifies trusted time servers

- Key rule: Always upgrade to the secure version of any protocol.